# MATH 2265 – R Group Project

In this project, you will work in a **group of two students**. Please read all instructions carefully before starting and before submitting your assignment.

## Group Submission Rules

1. Submit only ONE file per group (not one per student).
2. Save your file name exactly in this format: First.Student_Second.Student
   Example: Hani.Aldirawi_James.Smith.pdf
3. Your submission must include:
   - R code
   - R output/results (tables, numerical results, and graphs)

## Dataset Requirements

You must find a dataset that includes:

1. At least 30 observations (rows of data)
2. At least 2 numerical variables
3. At least 1 categorical variable

## Project Requirements

### 1. Data Description (Short Paragraph)

Write 4–6 clear sentences explaining:

- What or who your data describes
- The number of observations
- The two numerical variables (and what they measure)
- The one categorical variable (and what it represents)
- Why you chose this dataset
- A link to the data source, or a short description of how you collected the data

**2. Numerical Data Analysis**

For the Two Numerical Variables, use **R** to report the following

- Mean
- Median
- Standard deviation
- Five-number summary (min, Q1, median, Q3, max)
- Create one Histogram (with title and labeled axes) for each variable
- Create one Box Plot (with title and labeled axes)

**Interpretation:** Write **3–4 sentences** interpreting your numerical results and graphs.

**3. Categorical Data Analysis**

For the One Categorical Variable, use **R** to report the following:

- Frequency table (counts)
- Create one Pie Chart
- Create one Bar Graph

Suggestion: Below are some links to help you find data

1. Excel Sample Data for Practice or Training Example
2. Datasets/Raw Data - Health Statistics & Data - Library Guides at UC Berkeley
3. Search Results - data.gov.uk
4. Global Health Data Exchange | GHDx
5. https://www.baseball-reference.com/
6. Find Open Datasets and Machine Learning Projects | Kaggle
7. Any financial website.

**R Group Project Rubric**

| Criterion | Requirements | Points |
|---|---|---|
| 1. Group participation | The project is completed by a group of exactly two students. Both students contribute to the project. | 5 |
| 2. Names, filename, and submission | Both students' names are listed correctly; the filename follows First.Student_Second.Student.pdf; only one file is submitted for the group. | 5 |
| 3. R code and output | The submission includes the R code and corresponding R output, including numerical results, tables, and graphs. The code produces the submitted results. | 15 |
| 4. Data description | The dataset meets all requirements and the 4–6 sentence description includes the required information. | 20 |
| 5. First numerical variable | Complete and accurate analysis of the first numerical variable. | 10 |
| 6. Second numerical variable | Complete and accurate analysis of the second numerical variable. | 10 |
| 7. Interpretation of numerical results | Clear interpretation of the numerical summaries, histograms, and box plots. | 10 |
| 8. Categorical variable analysis | Complete and accurate analysis of the categorical variable. | 15 |
| 9. Organization and presentation | The report is clear, complete, readable, and professionally organized. | 10 |
| **Total** | | **100** |